

Wind Turbine EDA: Where to Invest & What to Look For

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- **Wind power is booming:** The renewable energy market is growing really fast, and wind power has become one of the main ways the U.S. is getting its new energy.
- **Investors are interested, but it's tricky:** Big investment firms are buying up wind farms for the long run. However, no two sites are the same—things like the type of turbine, the size of the project, and the location all change how well it performs.
- **What this project does:** To help investors make smart and safe buying decisions, this project looks closely at data from over 70,000 wind turbines across the U.S.

Introduction & Background





Problem Statement & Goals

A private equity firm wants to acquire U.S. wind farms, but needs to know which turbine specs and locations perform best — before committing capital.



Which regions perform strongest?



How do turbine size and rotor size relate to performance?



Are there rare, high-performing turbine models worth a closer look?

Data & Methodology



Main Dataset

U.S. Wind Turbine Database (USWTDB) — 70,000+ turbines: location, capacity, hub height, rotor diameter, manufacturer.

The Challenge

Real Net Generation (MWh) data wasn't available.

Our Approach

Estimated Net Generation using: capacity x hours/year x a capacity factor scaled to each state's wind strength (20%-45%).

Data Cleaning



- Dropped unused ID/reference columns (FAA, USGS, image-source fields)
- Removed rows missing critical fields (project capacity, project year)
- Standardized manufacturer names (merged Siemens/Gamesa/Goldwind variants)
- Mapped state names to abbreviations across all datasets, for merging
- Built a Census Region column (Northeast/Midwest/South/West) from state



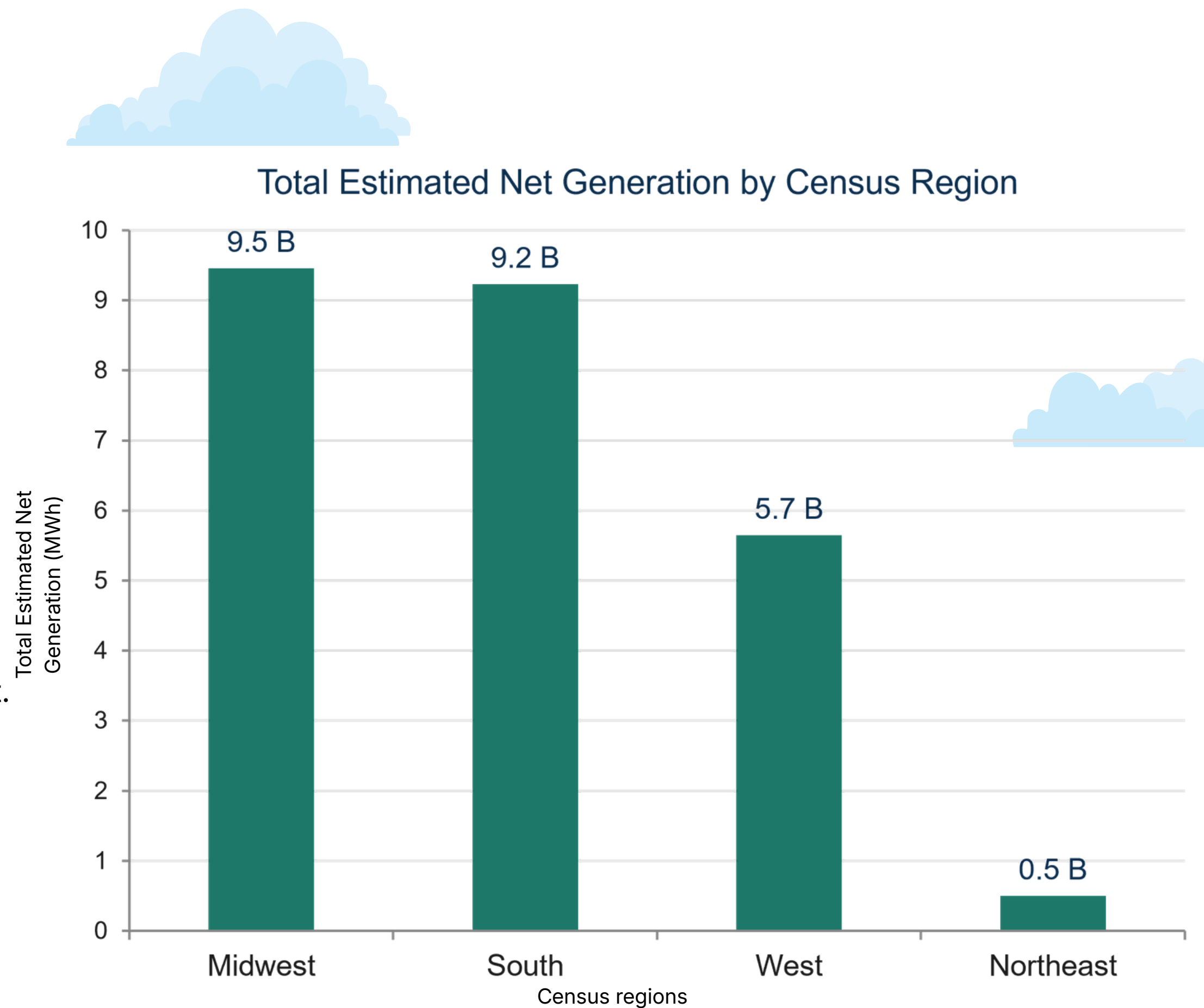
Deep Dive: Regional Performance

Midwest & South = ~75%
of total estimated output combined.

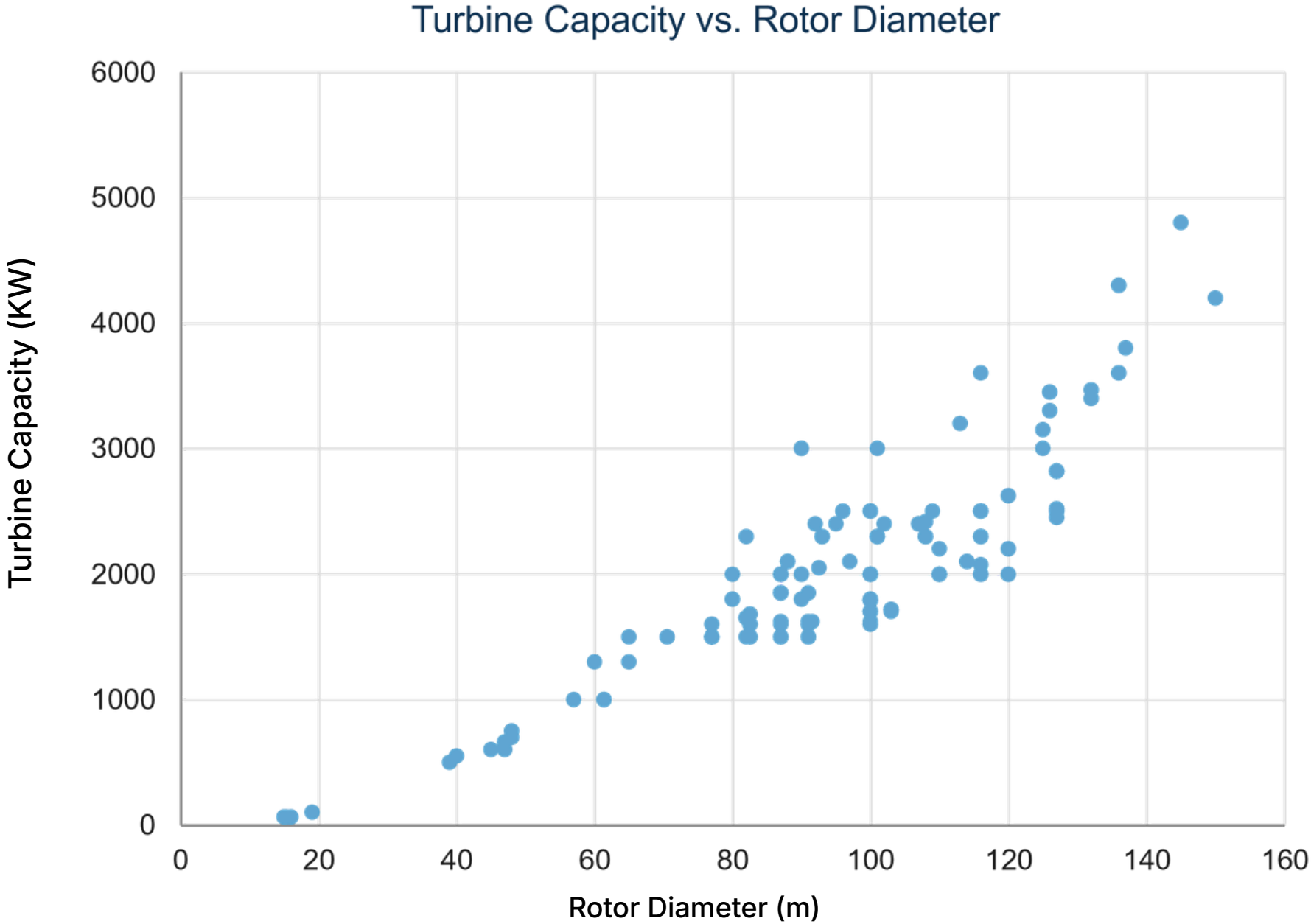
Midwest has the largest overall turbine count (25,106 turbines) and drives the highest total.

South has the strongest average performance per project.

Northeast lowest on every measure



Deep Dive: Turbine Specifications



Benchmark profile (top 10% performers):

- Rated capacity: ~2,378 kW
 - Hub height: ~86.5 m
 - Rotor diameter: ~111.6 m

clear pattern

bigger rotors = higher capacity, consistently.



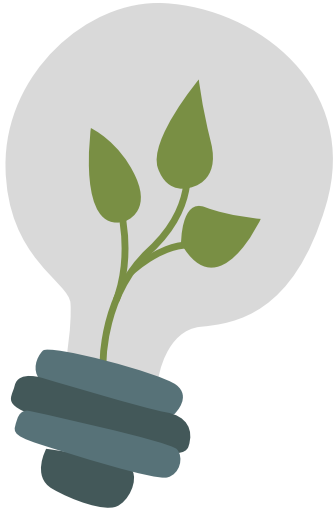
Deep Dive: Niche Opportunities



A few rare turbine models significantly outperform the fleet average despite low unit counts:

Model	Manufacturer	Units	Avg. Est. Net Gen (MWh)
AW132/3300	Nordex	55	888,599
V136-3.7	Vestas	29	797,413
AW132/3465	Nordex	79	688,204
GE 1.85-87	GE Wind	3	613,023
GE1.6-77	GE Wind	63	612,573
SG-2.6-114	Siemens	12	598,555





Summary of Findings



Focus on the Midwest and South: These two areas produce about 75% of all the energy, so they are definitely the most important regions.

Finding the best turbines: We created a standard profile to easily spot which wind turbines are doing the best job.

A few specific models stand out: A small number of specific turbine models do way better than the average turbine.

The Northeast is lagging behind: The Northeast region has the worst performance in every single category.

Recommendations



Focus on the Midwest and South first — they produce the most output.

Look for turbines have 100+m rotor diameter.

These numbers are estimates. Compare them to real data (like EIA-923) before deciding.

